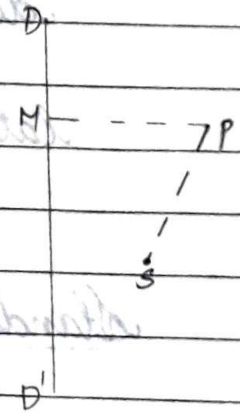
Date 12 / 11 / 2019

Analytic Geometry

Conic Section

The locus (path) of a point which moves in a plane so that its distance from a fixed point in the plane bears a constant ratio to its distance from a fixed straight line in the plane is called a conic section. The fixed point is called the focus the fixed line is called the directrix and the constant ratio is called eccentricity (e)

$$e = \frac{SP}{PM}$$



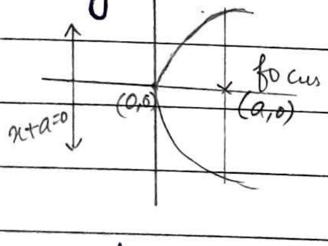
If $e = 1$: parabola

$e < 1$: ellipse

$e > 1$: hyperbola

Standard equation of parabola

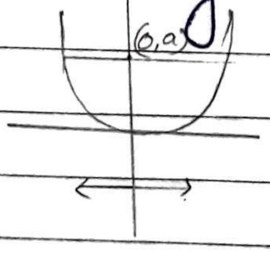
$$y^2 = 4ax$$



focus $\Rightarrow (a,0)$ lactus rectum:

directrix $\Rightarrow x + a = 0$

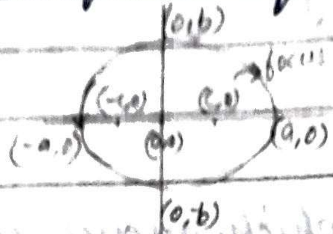
$$x^2 = 4ay$$



focus $\Rightarrow (0,a)$

directrix $\Rightarrow y + a = 0$

Standard equation of ellipse



$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Where $c = ae$

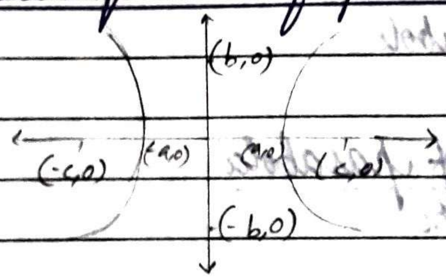
focus $\rightarrow (\pm ae, 0)$

Direction $\rightarrow \pm \frac{a}{e} = x$

latis rectum $\rightarrow \frac{2b^2}{a}$

$a^2 = b^2 + c^2$

Standard equation of parabola



$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

$$b^2 = c^2 - a^2$$

focus $\rightarrow (\pm ae, 0)$

direction $\rightarrow$ $x = \pm\frac{a}{e}$

latis rectum $= \frac{2b^2}{a}$